

IVÁN MANUEL PARRAS BERROCAL, PhD.

PhD in Marine Science and Technology

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SUMMARY

Oceanographer and Climate Scientist with 6+ years of experience in climate modeling (CMIP5/CMIP6), satellite altimetry (DUACS), reanalysis (CMEMS, ERA5), and scientific programming (Python, MATLAB). Expert in large-scale geospatial data analysis and climate projections, with a strong focus on ocean circulation, sea-level change, and climate impacts in the Mediterranean Sea.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher

CNRS/Météo-France, Toulouse (France) | Jun 2023 - Present

- Led an international collaboration across 10 institutes, assembling 28 simulations from 9 regional climate system models (RCSMs).
- Built standardized pipelines and produced the largest regional, multi-decadal CMIP5/CMIP6 ensemble to date for the Mediterranean Sea.
- Validated climate model simulations using satellite altimetry (e.g., DUACS) and Copernicus Marine Service reanalysis, enforcing cross-model comparability and quality control.
- Quantified RCP/SSP climate risks (sea level, circulation, warming) to inform stakeholder decision-making; authored 2+ peer-reviewed (Q1) papers and 6 manuscripts under review in Q1 journals, and presented findings at 5+ international conferences.

Predoctoral Researcher

Applied Physics Department, University of Cádiz (Spain) | Jun 2020 – Jun 2023

- Processed large oceanographic and atmospheric datasets (models, reanalysis, satellite observations) with advanced statistical methods to uncover correlations, trends, and biases.
- Participated in and organized oceanographic research cruises, contributing to the collection and analysis of critical marine data.

Graduate Research Fellow

Applied Physics Department, University of Cádiz (Spain) | Jan 2018 – Jul 2019

- Developed and calibrated hydrodynamic/water-quality models for hydrographic basins, integrating multi-source GIS layers (topography, land use, discharge).

EDUCATION

Ph.D., Marine Science and Technology, University of Cádiz (Spain) | 2020-2023

Thesis: *Study of the future climate evolution of the Mediterranean region with a regionally coupled climate model* | Awards: International Distinction; Highest Honors (*summa cum laude*)

M.Sc., Oceanography, University of Cádiz (Spain) | 2016-2017

Awards: Best Academic Record

B.Sc., Marine Sciences, University of Cádiz (Spain) | 2012-2016

SKILLS

Languages: Spanish (Native), English (Professional) and French (Pre-intermediate).

Programming & Tools: Python (xarray, NumPy, pandas, Matplotlib, scikit-learn, GDAL); MATLAB; CDO; NCO, GIS applications; Linux; Bash/shell scripting; HPC (high-performance computing).

Data and modelling: Climate modeling (CMIP5/CMIP6); scenario projections (RCP/SSP); geospatial data management (netCDF, GRIB); satellite altimetry (DUACS, SWOT); reanalysis (CMEMS, ERA5); oceanographic instrumentation (CTD, hydrographic sensors).

PEER-REVIEWED PUBLICATIONS

10. **Parras-Berrocal, I.M.**, Waldman, R., Gonzalez, N., Ahrens, B., Cabos, W., Jordà, G., Lionello, P., Sannino, G., Somot, S. (2025). Regionalized Mediterranean relative sea level projections under high-emission regional climate scenarios. *Environmental Research Letters*, 20, 114068, <https://doi.org/10.1088/1748-9326/ae15a5>
9. **Parras-Berrocal, I.M.**, Waldman, R., Sevault, F., Somot, S., Gonzalez, N., Ahrens, B., et al. (2024). Response of the Mediterranean Sea surface circulation at various global warming levels: A multi-model approach. *Geophysical Research Letters*, 51, e2024GL111695. <https://doi.org/10.1029/2024GL111695>
8. Vázquez, R., **Parras-Berrocal, I.M.**, Cabos, W., Sein, D. Mañanes, R., Bolado-Penagos, M. & Izquierdo, A. (2024). Climate change in the Canary/Iberia upwelling region: the role of ocean stratification and wind. *Environmental Research Letters*, 074064. <https://doi.org/10.1088/1748-9326/ad5ab4>
7. **Parras-Berrocal, I. M.**, Vazquez, R., Cabos, W., Sein, D., Álvarez, O., Bruno, M. & Izquierdo, A. (2023). Dense water formation in the Eastern Mediterranean under global warming scenario. *Ocean Science*, 19(3), 941-952. <https://doi.org/10.5194/os-19-941-2023>
6. Vázquez, R., **Parras-Berrocal, I.M.**, Koseki, S., Cabos, W., Sein, D. & Izquierdo, A. (2023). Seasonality of coastal upwelling trends in the Mauritania-Senegalese region under RCP8.5 climate change scenario. *Science of the Total Environment*, 898, 166391. <https://doi.org/10.1016/j.scitotenv.2023.166391>
5. **Parras-Berrocal, I. M.**, Vazquez, R., Cabos, W., Sein, D., Álvarez, O., Bruno, M. & Izquierdo, A. (2022). Surface and Intermediate Water Changes Triggering the Future Collapse of Deep Water Formation in the North Western Mediterranean. *Geophysical Research Letters*, 49. <https://doi.org/10.1029/2021GL095404>
4. de la Vara, A., **Parras-Berrocal, I. M.**, Izquierdo, A., Sein, D. V., & Cabos, W. (2022). Climate change signal in the ocean circulation of the Tyrrhenian Sea, *Earth System Dynamics*, 13, 303–319. <https://doi.org/10.5194/esd-13-303-2022>
3. Vazquez, R., **Parras-Berrocal, I.**, Cabos, W., Sein, D. V., Mañanes, R. & Izquierdo, A. (2022). Assessment of the Canary current upwelling system in a regionally coupled climate model. *Climate Dynamics* 58, 69–85. <https://doi.org/10.1007/s00382-021-05890-x>
2. **Parras-Berrocal, I. M.**, Vazquez, R., Cabos, W., Sein, D., Mañanes, R., Perez-Sanz, J., & Izquierdo, A. (2020). The climate change signal in the Mediterranean Sea in a regionally coupled atmosphere–ocean model, *Ocean Science*, 16, 743–765. <https://doi.org/10.5194/os-16-743-2020>
1. Roque, D., **Parras-Berrocal, I.**, Bruno, M., Sánchez-Leal, R. & Hernández-Molina, F. J. (2019). Seasonal variability of intermediate water masses in the Gulf of Cadiz. Implications of the Antarctic and Subarctic seesaw model, *Ocean Sci.*, 15, 1381-1397. <https://doi.org/10.5194/os-15-1381-2019>

BOOK CHAPTERS

2. Vázquez, R., **Parras-Berrocal, I.**, Cabos, W., Sein, D., Mañanes, R. Pérez Sanz J.I & Izquierdo, A. (2019). Climate Evaluation of a High-Resolution Regional Model over the Canary Current Upwelling System. *Computational Science – ICCS 2019*, Springer Nature, 240-252
1. **Parras-Berrocal, I.M.**, Cabos, W., Izquierdo, A., Vázquez, R., Mañanes, R. & Sein, D. (2018). Señal del cambio climático en el mar Mediterráneo: perspectiva de un modelo climático regionalmente acoplado atmósfera-océano. *Publicaciones de la Asociación Española de Climatología (AEC)*, 11, 827-836. <http://hdl.handle.net/20.500.11765/9968>

EXPERT ASSESSMENT REPORTS

1. Vargas-Yáñez, M., Marcos, M., Fontela, M., & Somavilla, R., et al. (2024). Capítulo 4. Cambios

Físicos y Biogeoquímicos en el Océano alrededor de España durante el periodo observacional: variabilidad, tendencias y forzamientos. Informe CLIVAR-SPAIN. Sobre el clima en España (187–230). Ministerio para la Transición Ecológica y el Reto Demográfico (MITECO). ISBN (en línea): 978-84-18778-47-6

SELECTED CONFERENCES

- **Parras-Berrocal I. M.**, Somot, S. and Waldman, R. (2026). Multi-model projections of marine climatic impact-drivers in the Mediterranean Sea under increasing global warming levels. Ocean Science Meeting 2026 (Glasgow, United Kingdom).
 - **Invited lecture, Parras-Berrocal I.M.** Understanding Mediterranean Surface Circulation through CMEMS Products: A Multi-Model Assessment. *Marine Data 4 the Mediterranean Sea 2025 (Workshop)*, Copernicus Marine Service, [16/09/2025]. Enlace: <https://events.marine.copernicus.eu/marine-data-4-mediterranean-sea>
 - **Parras-Berrocal I. M.** Waldman, R., Gonzalez, N. M., and Somot, S. (2025). Future projections of sea level rise in the Mediterranean Sea. EGU General Assembly, session CL 4.17 (Vienna, Austria).
 - **Parras-Berrocal, I. M.**, Waldman, R., Sevault, F., Gonzalez, N., & Somot, S. (2024). Dynamic response of the Mediterranean Sea surface circulation at various global warming levels: A multi-model approach. Ocean Science. EGU General Assembly (Vienna, Austria).
 - **Parras-Berrocal, I.M.**, Vazquez, R., Cabos, W., Sein, D., Álvarez, O., Bruno, M. & Izquierdo, A. (2023). Dense water formation in the Eastern Mediterranean under global warming scenario. Ocean Science. EGU General Assembly (Vienna, Austria).
- Summary:** 20+ national/international conference presentations. **Full record:** ORCID 0000-0003-4659-3924.

PROJECTS

- **ANR-22-P0CE-0003 (MEDIATION)**, Methodological developments for a robust and efficient digital twin of the ocean. *Postdoctoral researcher, 01/2025-ongoing.*
- **ANR-21-CE02-0017 (PopNCo)**, Evaluate future trends in Gorgonian connectivity in climate scenarios of the Mediterranean Sea. *Postdoctoral researcher, 06/2023-01/2025.*
- **RTI2018-100865-B-C22 (TRUCO)**, Circulation and transport processes in the estuaries of the gulf of Cadiz: current situation and projections of future climate change scenarios. *Predoctoral researcher, 06/2020-06/2023.*

RESEARCH STAYS

Visiting scientist, Physical Oceanography Group - University of Málaga (Málaga, Spain) 14/04/2025-18/08/2025 and 12/11/2025-28/02/2026: Intercomparison of Very-High Resolution Global, Global and Regional Climate models in the Mediterranean Sea.

Visiting scientist, Climate Dynamic section of Alfred Wegener Institute for Polar and Marine Research (Bremerhaven, Germany) 01/02/2022-30/04/2022: Assessment of the ability of a global ocean model to reproduce the observed regional climate of the Mediterranean Sea. Evaluation of the projected climate change signal in the Mediterranean Sea under CMIP5 and CMIP6 scenarios.

Visiting scientist, Department of Physics and Mathematics (University of Alcala, Madrid) 18/04/2021-23/04/2021: Assessing the impact of the collapse of the Northwest Mediterranean Deep Water Formation on changes in the characteristics of the Mediterranean outflow to the Atlantic.

Bachelor student, Oceanographic Institute of the Ecuadorian Navy (INOCAR, Galapagos Islands, Equator) 06/07/2016-06/08/2016: Development, implementation and improvement of a coupled hydro-dynamic-atmospheric model for tsunami prevention.

OCEANOGRAPHIC CAMPAIGNS

Research staff, Oceanographic campaigns of TRUCO Project - B.I.O UCádiz (Universidad de Cádiz) 09/2021. Current profiler, CTD, thermosalinograph, Niskin bottles and multihaz eco-Sound.

Research staff, ZEE 2019 - B.I.O Hespérides 06/2019. ADCP current profiler and XBT

Research staff, Oceanographic campaigns MIOUCA - B.I.O UCádiz (Universidad de Cádiz) 09/2018. Current profiler, CTD, thermosalinograph, Niskin bottles and multihaz eco-Sound.

Research staff, Oceanographic campaigns MIOUCA - B.I.O UCádiz (Universidad de Cádiz) 06/2018. Current profiler, CTD, thermosalinograph, Niskin bottles and multihaz eco-Sound.

Research staff, ZEE 2019 - B.I.O Hespérides 06/2019. ADCP current profiler, thermosalinograph and XBT.

COURSES, OTHER ACTIVITIES AND OUTREACH

2026

Scientific Committee Member, IX Encuentro de Oceanografía Física, Cádiz, Spain, 2026.

2025

Invited lecture, Oceanography Unveiled: Climate and oceans: from global to regional, University of Salento, Lecce (Italy), 1-5 September 2025

Animated GIFs from climate model outputs, Comment déjouer les incertitudes des modèles numériques ?, YouTube [L'Esprit Sorcier TV], [20/05/2025]. Link: [\[https://www.youtube.com/watch?v=0bUWsuIS-Cs\]](https://www.youtube.com/watch?v=0bUWsuIS-Cs)

2023

Intro to Python Course (25 hours). Teaching Innovation Program. University of Cadiz.

Introduction to the Weather Research and Forecasting (WRF) model (12 hours). University of Cadiz.

2021

Scientific dissemination Activities: European Researchers' Night (14 hours). University of Cadiz.

2020

Training Workshop for the North West Shelf Seas. Mercator Ocean International Event Team.

Training Workshop for the Mediterranean Sea. Mercator Ocean International Event Team.

Oceanographic mentor in Ocean Hackathon Cádiz. Ocean Hackathon.

Workshop Operational Oceanography (7.5 hours). The Centre for Marine and Environmental Research (CIMA) of University of Algarve.

Ecosystem Services in water management in Spain, Portugal and France (75 hours). FUECA. University of Cadiz.

2019

MATLAB Applied to Marine Sciences (20 hours). University of Cadiz Doctorate School (EDUCA).

Volunteer at the Fifth Euromarine General Assembly (10 hours).

TEACHING EXPERIENCE

Teaching collaboration in Marine Science Degree, Applied Physics Department. University of Cadiz | **Subject:** Applied physical oceanography, geophysical fluid mechanics, integrated dynamics of natural coast systems, geophysics and tectonics.

Teaching collaboration in Environmental Science Degree, Applied Physics Department. University of Cadiz | **Subject:** SIG and remote sensing, assessment of environmental pollution, climate change.

REVIEW

2023-ongoing, reviewer of journals: Ocean Science, Climate Dynamics, Scientific Reports, Ocean Modelling and Advances in Space Research.

REFeree CONTACT DETAILS

Dr. Samuel Somot, Postdoc supervisor | samuel.somot@meteo.fr

Dr. Robin Waldman, Postdoc supervisor | robin.waldman@meteo.fr

Dr. Alfredo Izquierdo, PhD supervisor | alfredo.izquierdo@uca.es